

Circuit Analysis (Lab # 2)

Experiment 2: Measurements of Waveforms and Fundamental Quantities, Basic resistive circuits.**Name: Muhammad Abdullah****Registration No: 2025-CE-54****Date: 03/15/2026****Grade and Signature:**

CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objectives: This experiment will provide exposure to measurements of waveforms and Fundamental Quantities, Basic resistive circuits. A primary purpose of this lab course is for you to master the use of electronic test equipment and measurement of fundamental quantities.

Equipment required:

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter.

Procedure**Experiment # 1****SIGNAL GENERATOR AND OSCILLOSCOPE**

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'A' shown below in Figure 1.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 100 Hz and peak-to-peak voltage of 5V.
4. Set up one probe across input signal, and another across R1 and R2 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

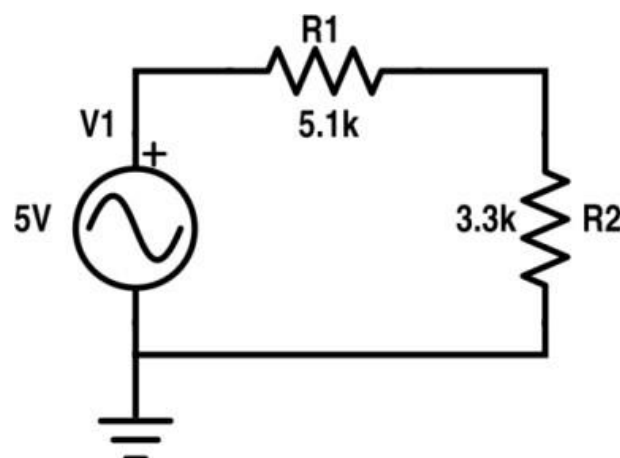


Figure 1 Circuit 'A'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	5 V	1.07 V	3.036 V
R2	5 V	0.694 V	1.964 V

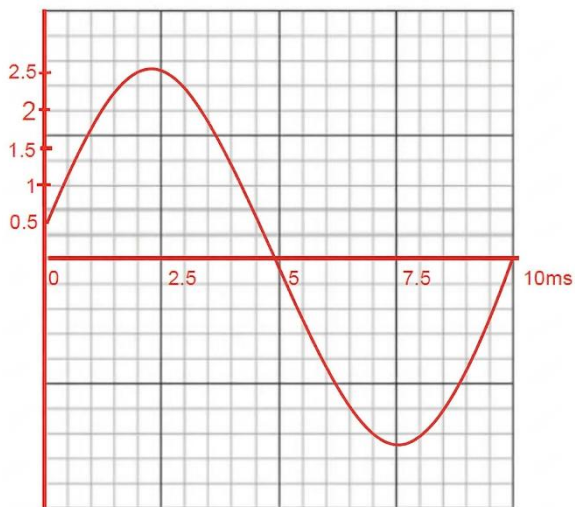


Figure 2 Input Voltage Signal

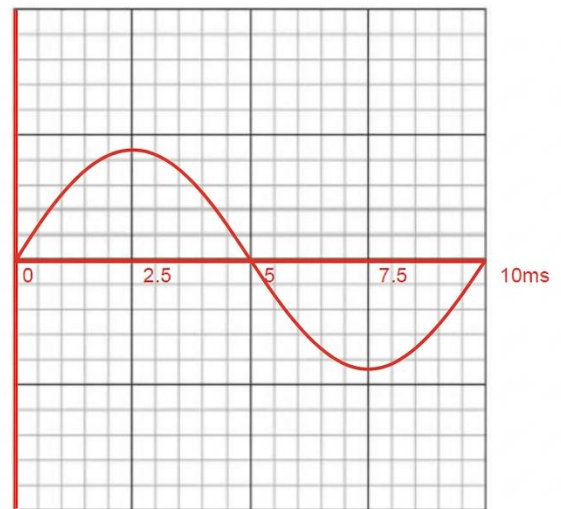


Figure 3 Voltage Signal across R1

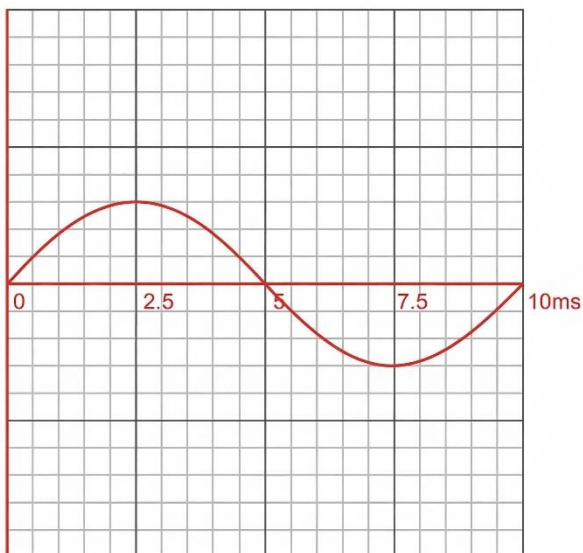


Figure 5 Voltage Signal across R2

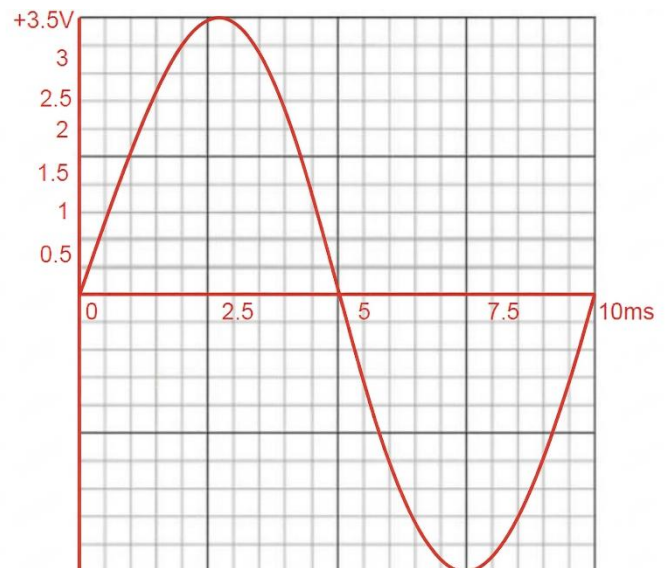


Figure 6 Draw 7V Sinusoidal

Experiment # 2

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'B' shown below in Figure 7.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 200 Hz and peak-to-peak voltage of 7V.
4. Set up one probe across input signal, and another across R1, R2 and R3 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

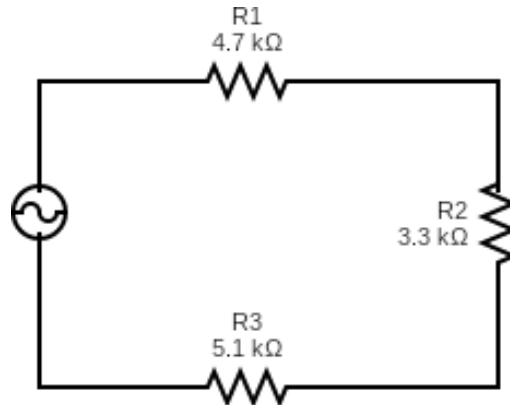


Figure 7 Circuit 'B'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	7 V	0.89 V	2.51 V
R2	7 V	0.62 V	1.76 V
R3	7 V	0.96 V	2.73 V
R1 and R2	7 V	1.51 V	4.27 V
R2 and R3	7 V	1.59 V	4.49 V

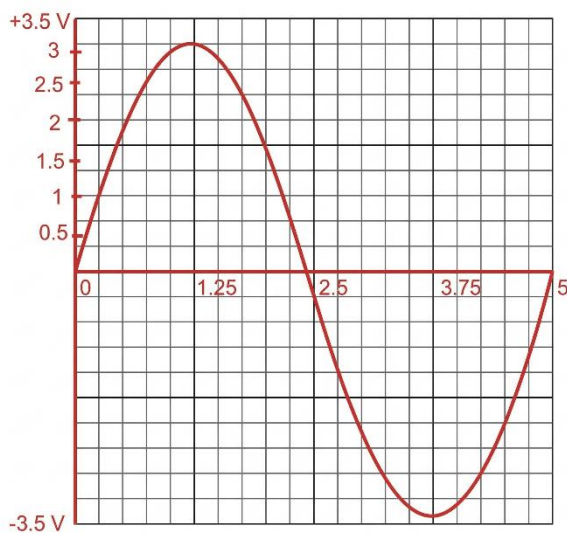


Figure 8 Input Voltage Signal

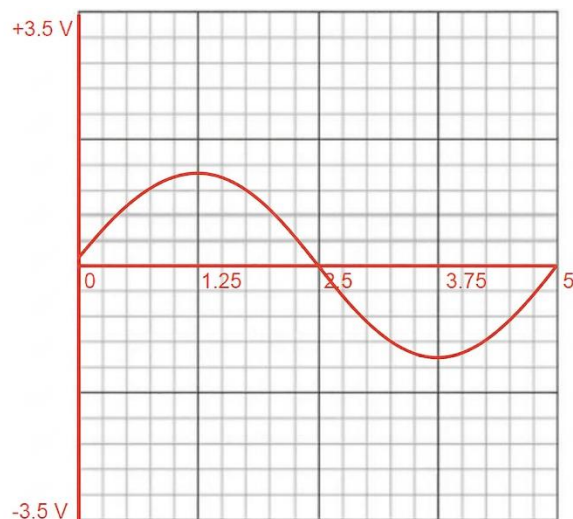


Figure 9 Voltage Signal across R1

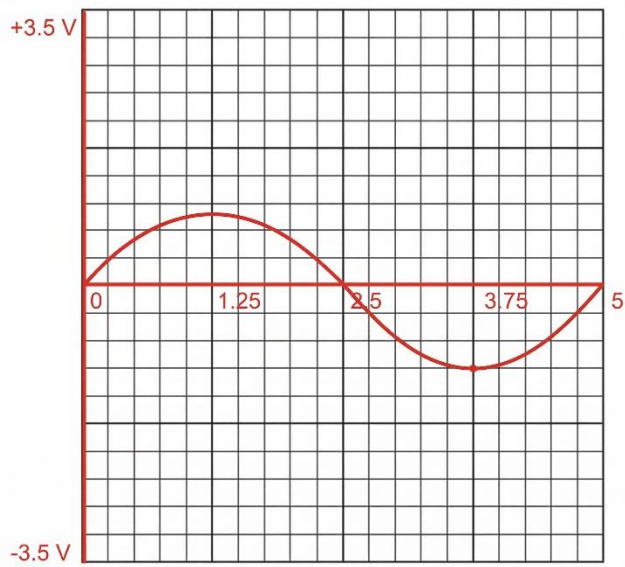


Figure 10 Voltage Signal across R2

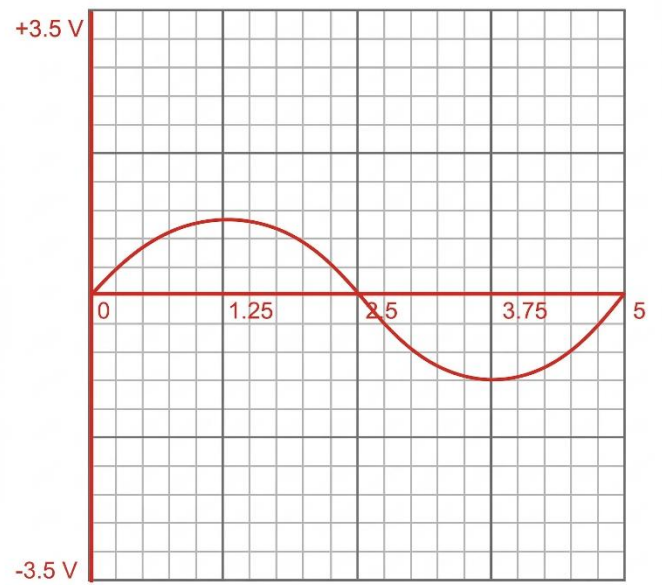


Figure 11 Voltage Signal across R3

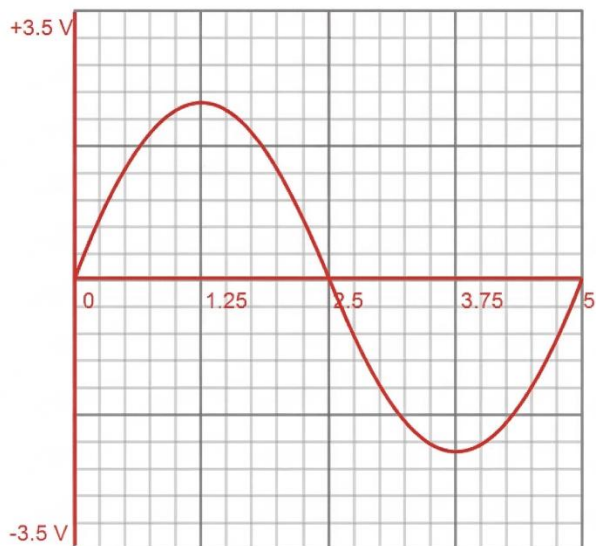


Figure 12 Voltage Signal across R1 and R2

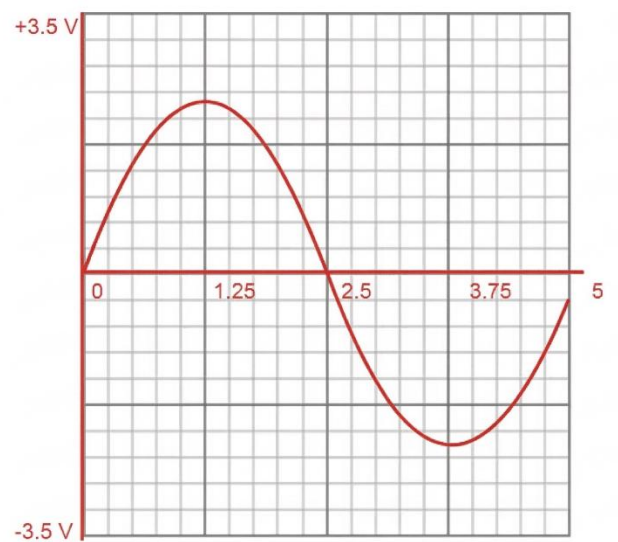


Figure 13 Voltage Signal across R2 and R3

Post Lab Questions

1. Define Ohm's Law.

Ohm's Law explains the relationship between voltage, current, and resistance in an electrical circuit. It states that the current flowing through a conductor is directly proportional to the voltage applied across it, as long as the temperature and other physical conditions remain constant. In simple words, when the voltage increases, the current also increases if the resistance does not change. The mathematical form of Ohm's Law is:

$$V = IR$$

In this equation, **V** represents voltage, **I** represents current, and **R** represents resistance measured in Ohms (Ω). Ohm's Law is one of the most important basic principles used in analyzing electrical circuits.

2. What is a resistor?

A resistor is a basic passive electronic component that is used to oppose or limit the flow of electric current in a circuit. It has two terminals and its main property is resistance, which is measured in Ohms (Ω). When electric current passes through a resistor, some of the electrical energy is converted into heat.

Resistors are commonly made using materials such as carbon, metal film, or wire-wound elements. In electronic circuits, resistors are used for many purposes such as controlling current, dividing voltage, adjusting signal levels, and protecting other components from excessive current.

3. How does a resistor work?

A resistor works by restricting the movement of electrons inside its material. When current flows through the resistor, electrons collide with the atoms present in the resistive material. Because of

these collisions, some electrical energy is lost in the form of heat. This process is known as Joule heating.

The power dissipated in a resistor can be written as:

$$P = I^2 R$$

The amount of resistance of a conductor depends on three factors: the resistivity of the material (ρ), the length of the conductor (L), and its cross-sectional area (A). The relationship is expressed as:

$$R = \rho L / A$$

If the resistance value is higher, the current flowing through the circuit will be smaller for the same applied voltage. This behavior follows the principle described by Ohm's Law.

4. What is your inference from the input and output waveforms on oscilloscope?

From the input and output waveforms observed on the oscilloscope, the following points can be concluded:

(a) The output waveforms measured across the resistors remain sinusoidal and have the same frequency as the input signal. This shows that resistors are linear components and they do not cause any phase shift or change in the signal frequency.

(b) The amplitude of the voltage across each resistor is smaller than the input voltage. This confirms the voltage divider behavior of the circuit. In Circuit A, the measured peak-to-peak voltages are approximately:

$$V_{R1} \approx 3.04 V_{pp}$$

$$V_{R2} \approx 1.96 V_{pp}$$

When these voltages are added together, the result is approximately 5 V, which verifies Kirchhoff's Voltage Law (KVL).

(c) In Circuit B, resistors R_2 and R_3 are connected in parallel. Because of this connection, both resistors

have the same voltage across them. The voltage across the parallel branch is smaller than the voltage across R1. This matches the expected result based on the equivalent resistance of the parallel combination compared with the total circuit resistance.

5. Show calculations of 7V input signal (Figure 6) from signal generator.

a. Voltage Calculations (using volt/div and number of divisions on oscilloscope)

Oscilloscope setting:

Volt/div = 1 V/div

Observed peak of waveform = 3.5 divisions above the center line

Peak Voltage:

$$V_{\text{peak}} = \text{divisions} \times \text{volt/div}$$

$$V_{\text{peak}} = 3.5 \times 1 = 3.5 \text{ V}$$

Peak-to-Peak Voltage:

$$V_{\text{pp}} = 2 \times V_{\text{peak}}$$

$$V_{\text{pp}} = 2 \times 3.5 = 7 \text{ V}$$

RMS Voltage:

$$V_{\text{rms}} = V_{\text{peak}} / \sqrt{2}$$

$$V_{\text{rms}} = 3.5 / 1.4142 \approx 2.475 \text{ V}$$

b. Frequency / Period Calculations (following frequency measurement steps)

Signal generator frequency:

$$f = 200 \text{ Hz}$$

Time period:

$$T = 1 / f$$

$$T = 1 / 200 = 0.005 \text{ s} = 5 \text{ ms}$$

Oscilloscope setting:

Time/div = 1 ms/div

One complete cycle occupies 5 divisions on the screen.

Verification:

$$T = 5 \text{ div} \times 1 \text{ ms/div} = 5 \text{ ms}$$

$$f = 1 / T$$

$$f = 1 / (5 \times 10^{-3}) = 200 \text{ Hz}$$

c. Current Calculation using Ohm's Law (Circuit B, 7V source)

Given:

$$R_1 = 4.7 \text{ k}\Omega$$

$$R_2 = 3.3 \text{ k}\Omega$$

$$R_3 = 5.1 \text{ k}\Omega$$

Step 1 — Equivalent resistance of parallel combination ($R_2 \parallel R_3$)

$$R_{eq} = (R_2 \times R_3) / (R_2 + R_3)$$

$$R_{eq} = (3300 \times 5100) / (3300 + 5100)$$

$$R_{eq} = 16,830,000 / 8400 \approx 2002.5 \Omega$$

Step 2 — Total resistance of the circuit

$$R_{total} = R_1 + R_{eq}$$

$$R_{total} = 4700 + 2002.5 = 6702.5 \Omega$$

Step 3 — Peak source voltage

$$V_{peak} = V_{pp} / 2$$

$$V_{peak} = 7 / 2 = 3.5 \text{ V}$$

Step 4 — Peak current

$$I_{peak} = V_{peak} / R_{total}$$

$$I_{peak} = 3.5 / 6702.5 \approx 0.5222 \text{ mA}$$

Step 5 — RMS current

$$I_{rms} = I_{peak} / \sqrt{2}$$

$$I_{rms} = 0.5222 / 1.4142 \approx 0.369 \text{ mA}$$

Step 6 — Current through each branch

Current through R_1 :

$$I_{R_1} = I_{rms} = 0.369 \text{ mA}$$

Voltage across parallel combination:

$$V_{eq} = I_{rms} \times R_{eq}$$

$$V_{eq} = 0.369 \times 2002.5 \approx 0.739 \text{ V (rms)}$$

Current through R_2 :

$$I_{R_2} = V_{eq} / R_2$$

$$I_{R_2} = 0.739 / 3300 \approx 0.224 \text{ mA (rms)}$$

Current through R_3 :

$$I_{R_3} = V_{eq} / R_3$$

$$I_{R_3} = 0.739 / 5100 \approx 0.145 \text{ mA (rms)}$$

Verification using Kirchhoff's Current Law:

$$I_{R_2} + I_{R_3} = 0.224 + 0.145 = 0.369 \text{ mA}$$

$$I_{R_2} + I_{R_3} = I_{R_1}$$

Hence, the current relationship satisfies KCL.